

Bayesian Modeling and Online Learning of Personalized and Time-Varying Treatment Effects in Non-Stationary Mobile Health Data

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Introduction

- In modern health studies, mobile health data collection and sensors enable **continuous, real-time tracking** of behavior and mental states at scale.
- Advancing Resilience and Coping (ARC) study: Intensive longitudinal ecological momentary assessment (EMA) + sensor data to quantify how **crisis response plan (CRP)** use impacts **suicidal ideation (SI)** in mental health.
- Challenges:** **Non-stationary** dynamics, **heterogeneous** population, and the need for fast, **real-time (online)** learning and uncertainty quantification.

► Motivating Example: Suicidal Ideation

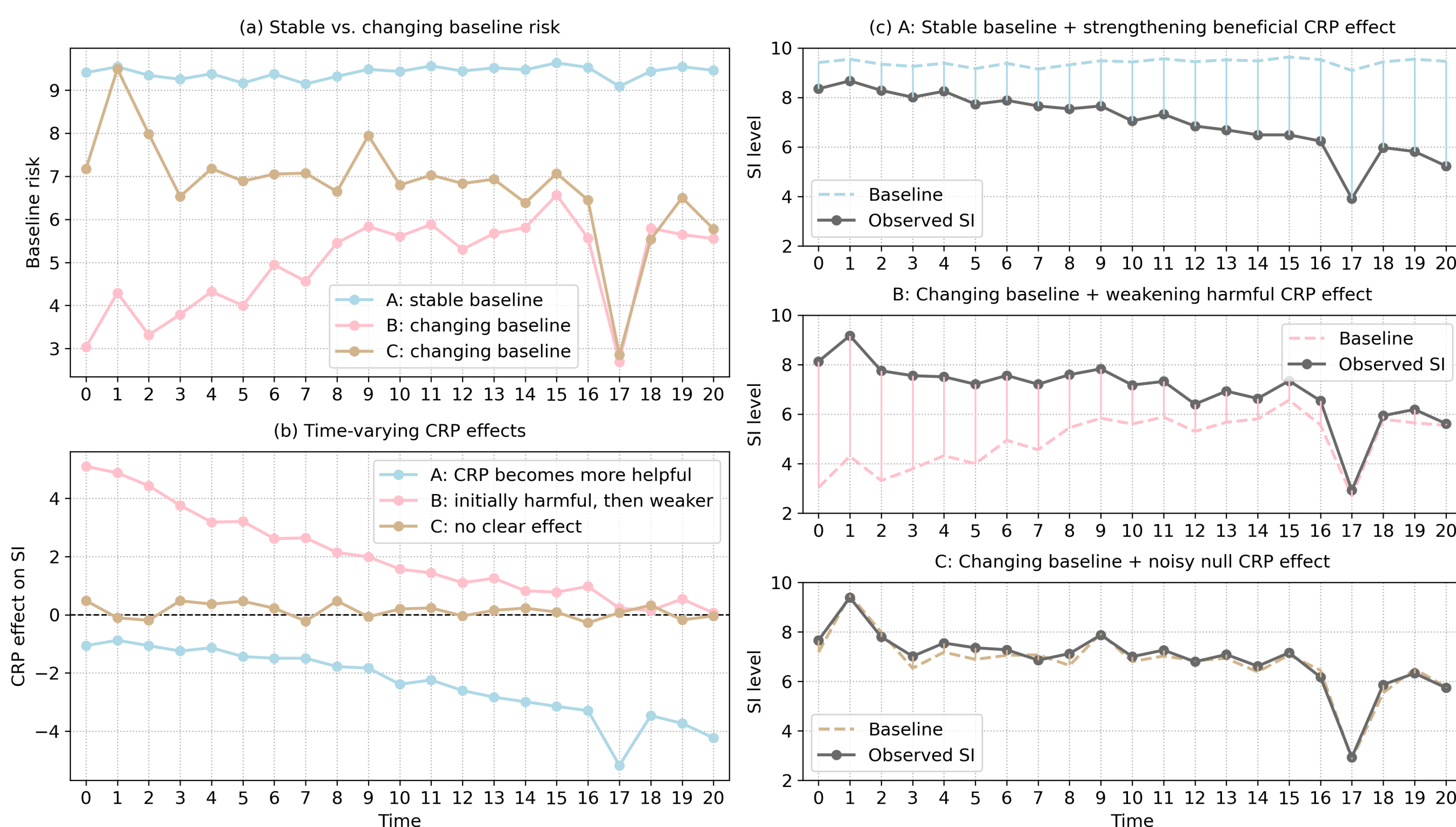


Figure 1. Superficially similar observed risk trajectories may arise from distinct underlying mechanisms, including differences in baseline risk dynamics and in the magnitude, direction, and timing of treatment effects.

- Heterogeneity in baseline risk and in the magnitude, direction, and timing of treatment effects **challenges population-averaged static models**, which can obscure clinically meaningful differences and limit individualized intervention (e.g., Fig. 1).
- Our solution: a **hierarchical Bayesian state-space framework** that
 1. **learns subject-specific effects**,
 2. **captures time-varying treatment dynamics**,
 3. **identifies latent subgroups** (harmful, null, beneficial),
 4. **enables short-term forecasting and sequential (online) updating**.

Conclusion

- We develop a **hierarchical Bayesian state-space framework** for **non-stationary mobile health data** that jointly captures **subject-specific baseline risk**, **personalized and time-varying treatment effects**, and **latent subgroup structure**.
- In simulations, the proposed approach achieves **accurate parameter recovery** and shows **clear improvement over GLMM** for forecasting future outcomes.
- The online learning component enables **sequential updating of cluster membership probabilities** for new patients **without refitting the model to the full data**, making the framework practical for real-time use.
- Substantively, this work supports **clinically actionable decision-making** by identifying not only who is likely to benefit from CRP use, but also when the intervention is most effective and when alternative intervention strategies may be more appropriate.
- More broadly, this work contributes to **precision medicine** by supporting **dynamic, individualized intervention strategies** that adapt to patients' changing risk profiles over time.

Methods

► Hierarchical Bayesian State-Space Framework

Key challenge: Data exhibit **time-varying effects**, **heterogeneity**, and require **real-time updating**.

We propose a hierarchical Bayesian state-space model that decomposes suicidal ideation severity into **baseline risk** and **dynamic treatment effects**. For individual i at time t ,

$$Y_{i,t} = \underbrace{\alpha_{i,t}}_{\text{baseline risk}} + \underbrace{\beta_{i,t}X_{i,t}}_{\text{time-varying effect} \times \text{treatment}} + \underbrace{\mathbf{C}_{i,t}^\top \boldsymbol{\eta}}_{\text{covariate adjustment}} + \underbrace{v_{i,t}}_{\text{noise}}, \quad v_{i,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_y^2).$$

Modeling non-stationarity: Subject-specific latent states $\boldsymbol{\theta}_{i,t} = (\alpha_{i,t}, \beta_{i,t})^\top$ evolve over time via random walks:

$$\alpha_{i,t} = \alpha_{i,t-1} + w_{i,t}^{(\alpha)}, \quad w_{i,t}^{(\alpha)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\alpha,i}^2); \quad \beta_{i,t} = \beta_{i,t-1} + w_{i,t}^{(\beta)}, \quad w_{i,t}^{(\beta)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\beta,i}^2).$$

Modeling heterogeneity: We introduce latent indicators $\mathbf{D}_i = (D_{0,i}, D_{1,i}, D_{\alpha,i}, D_{\beta,i})$ to capture two clinically important sources of heterogeneity. Specifically,

1. $(D_{0,i}, D_{1,i})$ characterize the **initial treatment effect direction**. They distinguish whether subject i shows **no effect**, a **harmful effect**, or a **beneficial effect** at time $t = 0$.

$$\beta_{i,0} \mid D_{0,i}, D_{1,i} \sim \begin{cases} \delta_0, & D_{0,i} = 0 \\ \mathcal{N}(-\mu_1, \tau_1^2), & D_{0,i} = 1, D_{1,i} = 1 \quad (\text{beneficial}), \\ \mathcal{N}(\mu_2, \tau_2^2), & D_{0,i} = 1, D_{1,i} = 0 \quad (\text{harmful}). \end{cases}$$

2. $(D_{\alpha,i}, D_{\beta,i})$ characterize the **temporal dynamics**. They determine whether the baseline risk trajectory $\alpha_{i,t}$ and treatment effect trajectory $\beta_{i,t}$ are **static** or **time-varying**.

$$\sigma_{\alpha,i}^2 \mid D_{\alpha,i} \sim \begin{cases} \delta_0, & D_{\alpha,i} = 0 \quad (\text{static}), \\ \text{Exp}(\lambda_\alpha), & D_{\alpha,i} = 1 \quad (\text{varying}), \end{cases} \quad \sigma_{\beta,i}^2 \mid D_{\beta,i} \sim \begin{cases} \delta_0, & D_{\beta,i} = 0 \quad (\text{static}), \\ \text{Exp}(\lambda_\beta), & D_{\beta,i} = 1 \quad (\text{varying}). \end{cases}$$

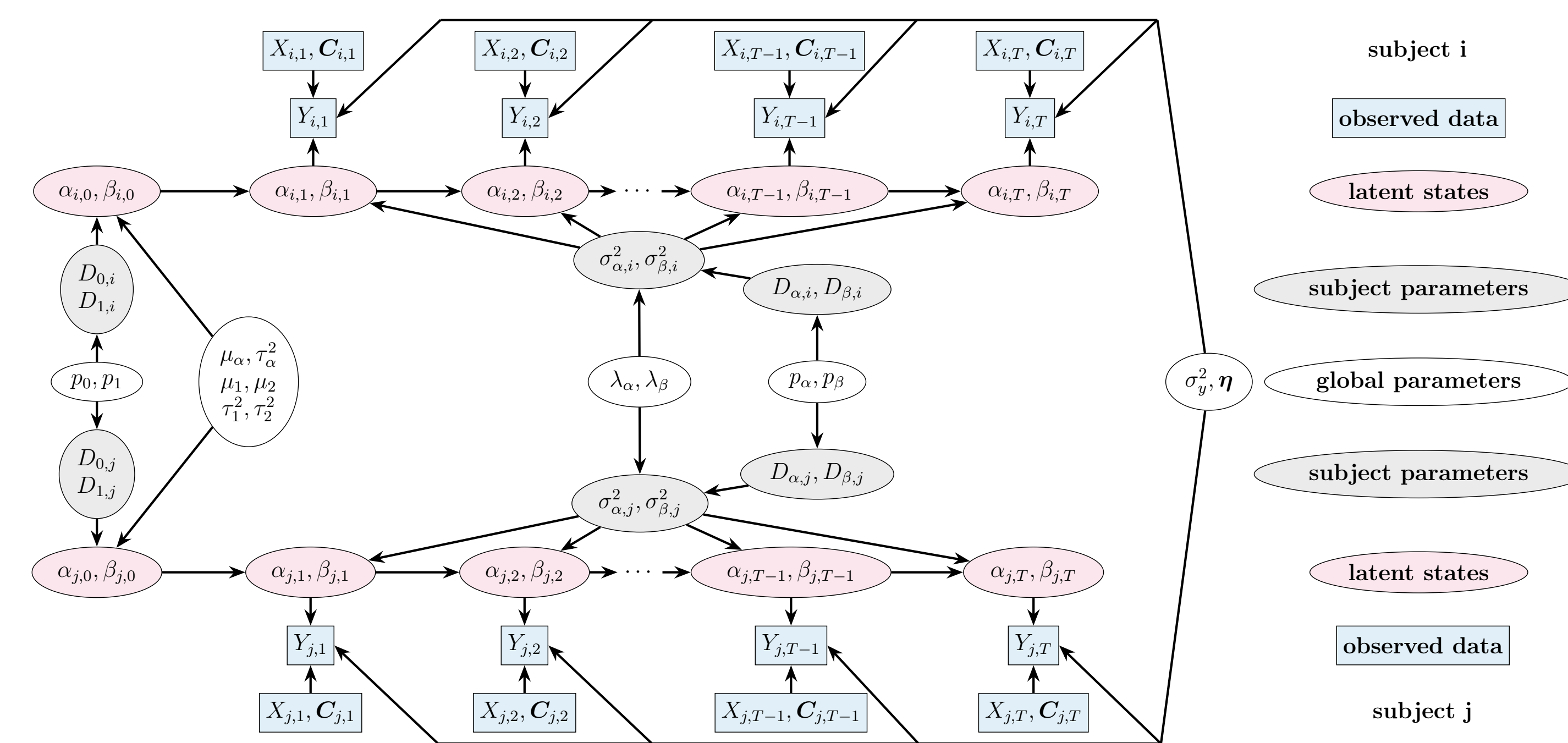


Figure 2. Graphical representation of the hierarchical Bayesian state-space model.

► Inference & Prediction

Posterior inference requires a hybrid MCMC framework integrating **FFBS** for dynamic latent states, **Gibbs updates** for high-dimensional continuous parameters, and **reversible-jump MCMC** for dimension-varying parameters defining the subgroup structure.

Q-step-ahead prediction: We develop a posterior predictive sampling algorithm that targets:

$$p(Y_{i,t+q} \mid \mathcal{H}^t, X_{i,t+q}, \mathbf{C}_{i,t+q}), \quad \text{where } \mathcal{H}^t = \{(Y_{i,h}, X_{i,h}, \mathbf{C}_{i,h}) : 1 \leq h \leq t, i = 1, \dots, N\},$$

which enables **short-term, individualized forecasting**.

Online learning: Sequentially update a new subject's subgroup membership (effect direction and dynamics), $\Pr(\mathbf{D}_{i^*} = \mathbf{d} \mid \mathcal{H}^t, \mathcal{H}_{i^*})$, where $\mathcal{H}_{i^*}$ denotes the new subject's history.

- Challenge:** Real-time inference for a new subject in a hierarchical, non-stationary model with latent subgroup structure.
- Approach: Sequential Monte Carlo (SMC)**
 1. Initialize particles using posterior samples from the fitted model and data-driven initialization of the new patient's individual parameters.
 2. Update particle weights as observations from new patients arrive sequentially.
 3. Apply resampling and model-based perturbation to preserve particle diversity.
- Innovation: real-time, subject-specific updating without refitting the model to the full data.**

Results

Simulation 1: Inference on model parameters

- Accurate estimate** of both global parameters and subject-specific latent trajectories.
- High classification accuracy** for subject-level indicators (> 90%).

Simulation 2: Q-step-ahead prediction

- Consistently outperforms GLMM** across all forecast horizons.
- Sharper predictions:** Tighter posterior predictive intervals and good coverage relative to GLMM (Fig. 3).

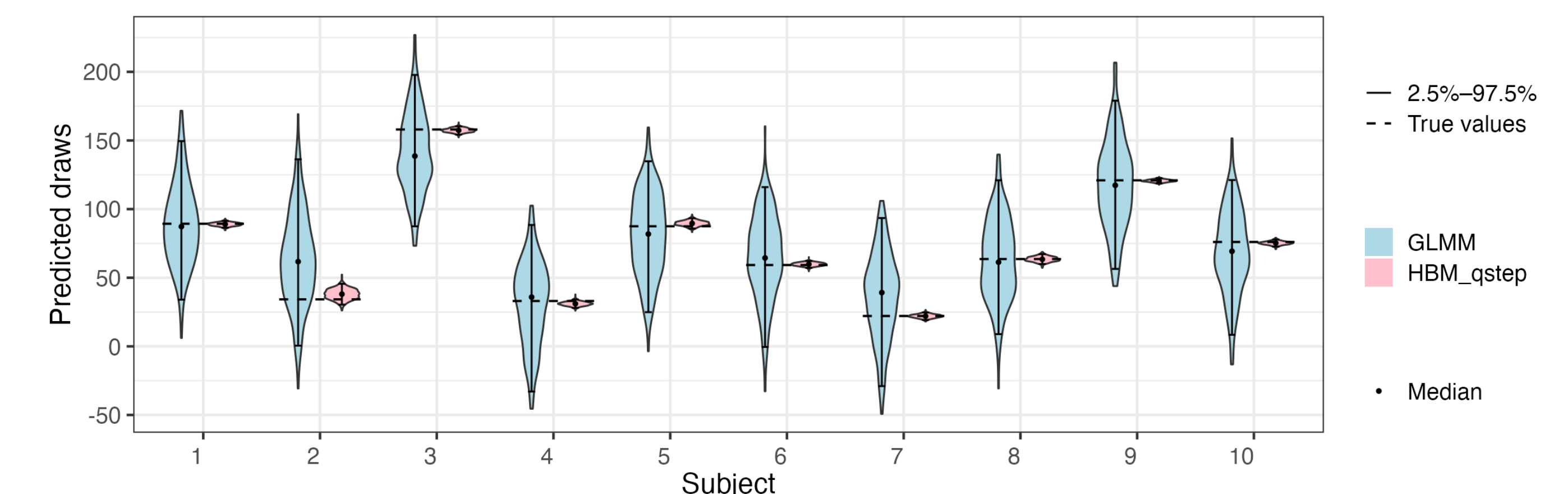


Figure 3. One-step-ahead posterior predictive distributions for 10 representative subjects under the hierarchical Bayesian model and the linear mixed model (GLMM) comparison.

Simulation 3: Online learning

- Matches batch MCMC:** SMC-based assignments align with full-batch model results.
- Uncertainty-aware:** Individuals misclassified by SMC are also challenging to classify under full batch MCMC (subject 101 in Fig. 4; subjects 116 and 119 in Fig. 5).
- Key gain:** Real-time updating without refitting the full model.

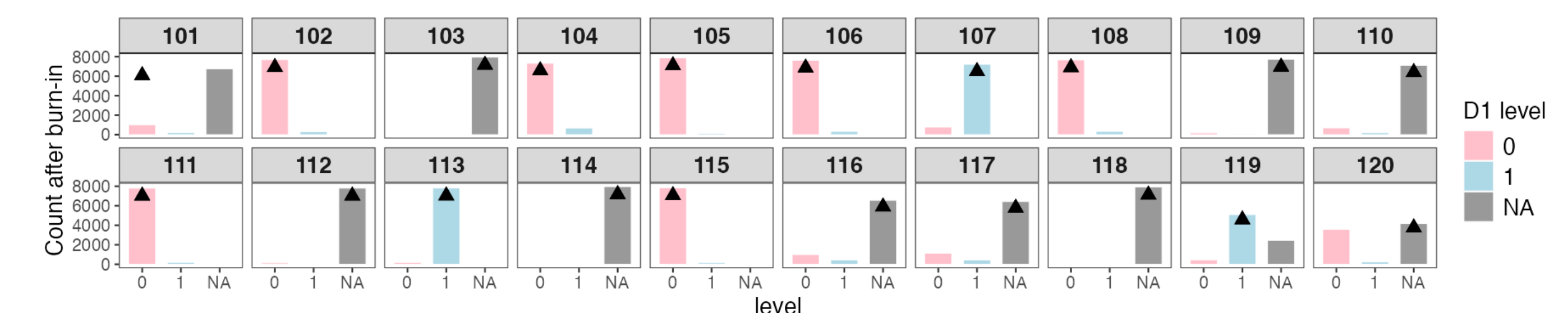


Figure 4. Posterior samples of $D_{1,i}$ (▲ = truth) for the 20 new subjects under the batch MCMC fit to all 120 subjects.

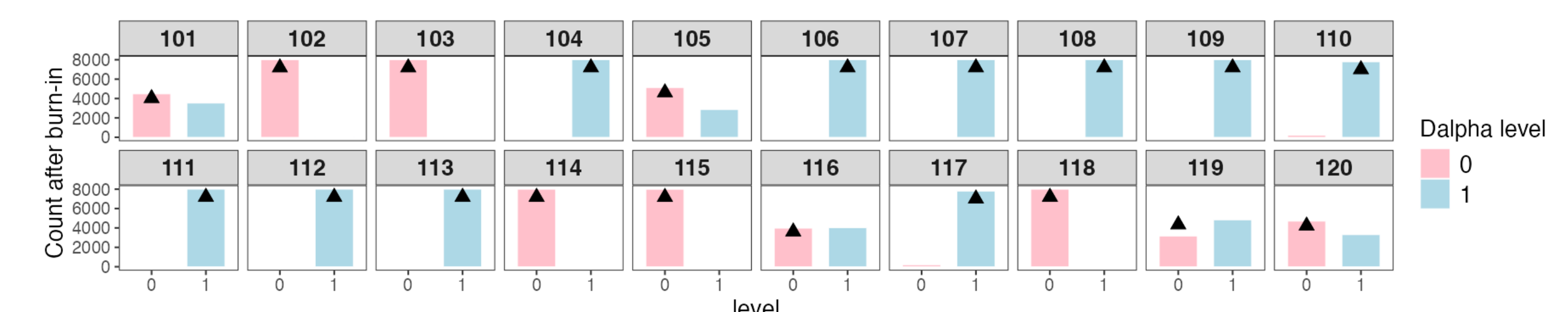


Figure 5. Posterior samples of $D_{\alpha,i}$ (▲ = truth) for the 20 new subjects under the batch MCMC fit to 120 subjects.